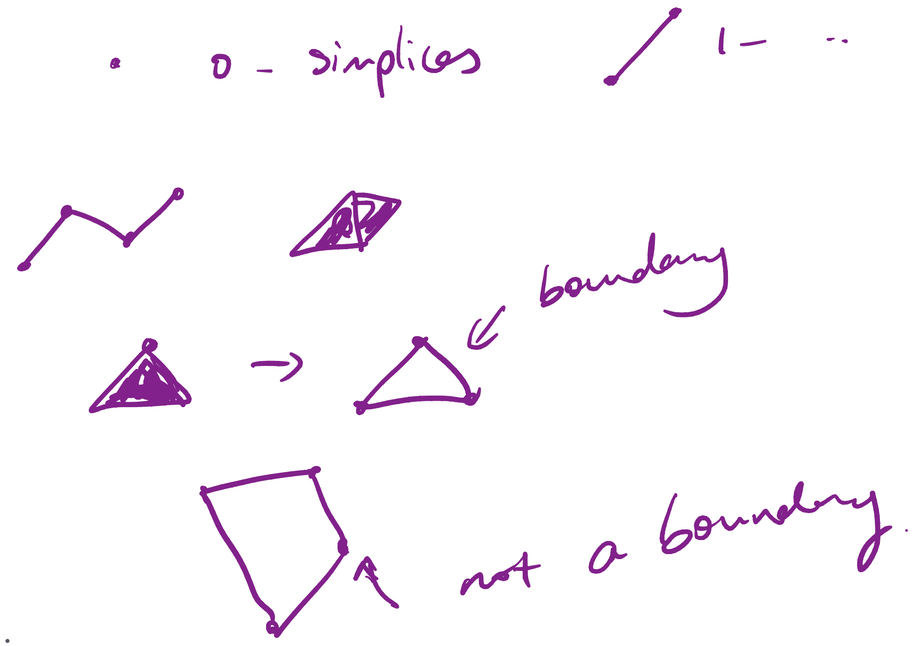
Filtrations and Induced Map

And maybe persistence?

Dr. Elena Xinyi Wang, October 14, 2026

Quick review

- We want to compute the “shape”: we turn “shape” into linear algebra; homology is the machinery
- p-simplex: basis vector
- p-chains: vectors
- Boundary map: matrix
- Cycles: kernel of a matrix
- Boundaries: image of a matrix
- Homology: cycles modulo trivial cycles

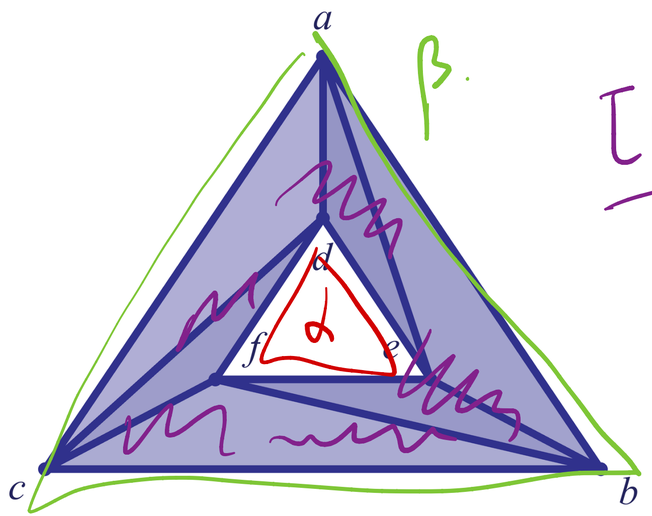


Homology!!

The pth homology group is the quotient space

$$H_p(K) := Z_p(K)/B_p(K)$$

[ ]

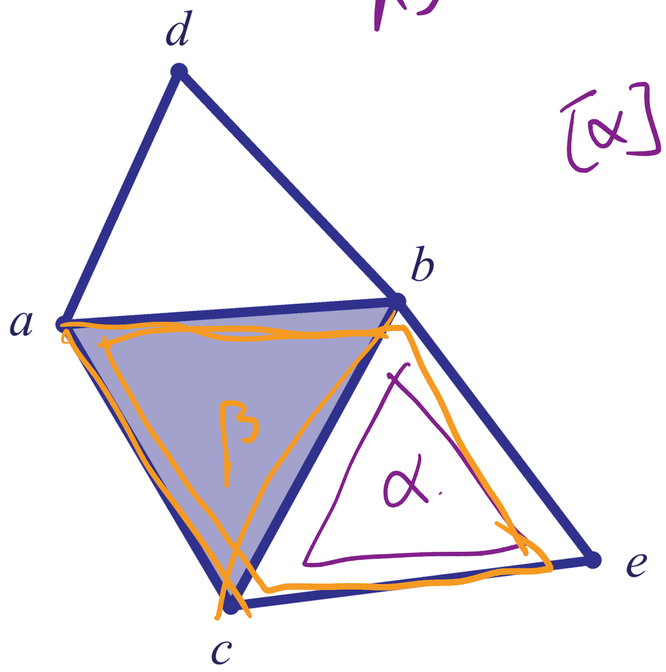


β.

[β] = [α].

AB + BE + EC + CA

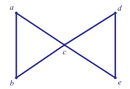
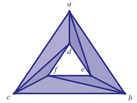
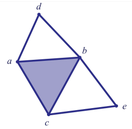
[α]

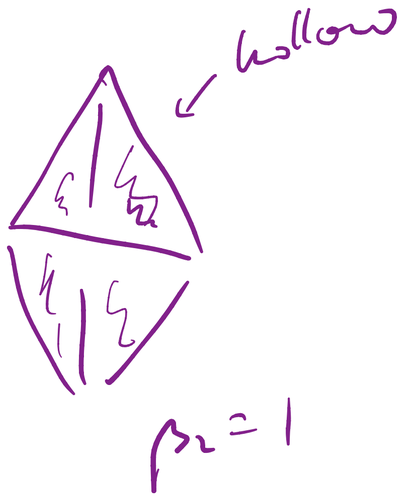


Betti Numbers

$$\beta_k = \dim H_k = \dim Z_k - \dim B_k$$

- β_0: number of connected components
- β_1: number of independent loop classes
- β_2: number of independent 2-dimensional homology classes

	β_0	β_1	β_2
	1	2	0
	1	1	0
	1	2	0



Simplicial Complex Filtration

A *filtration* $\mathcal{F} = \mathcal{F}(K)$ of a simplicial complex K is a nested sequence of its subcomplexes $\mathcal{F} : \emptyset = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_n = K.$

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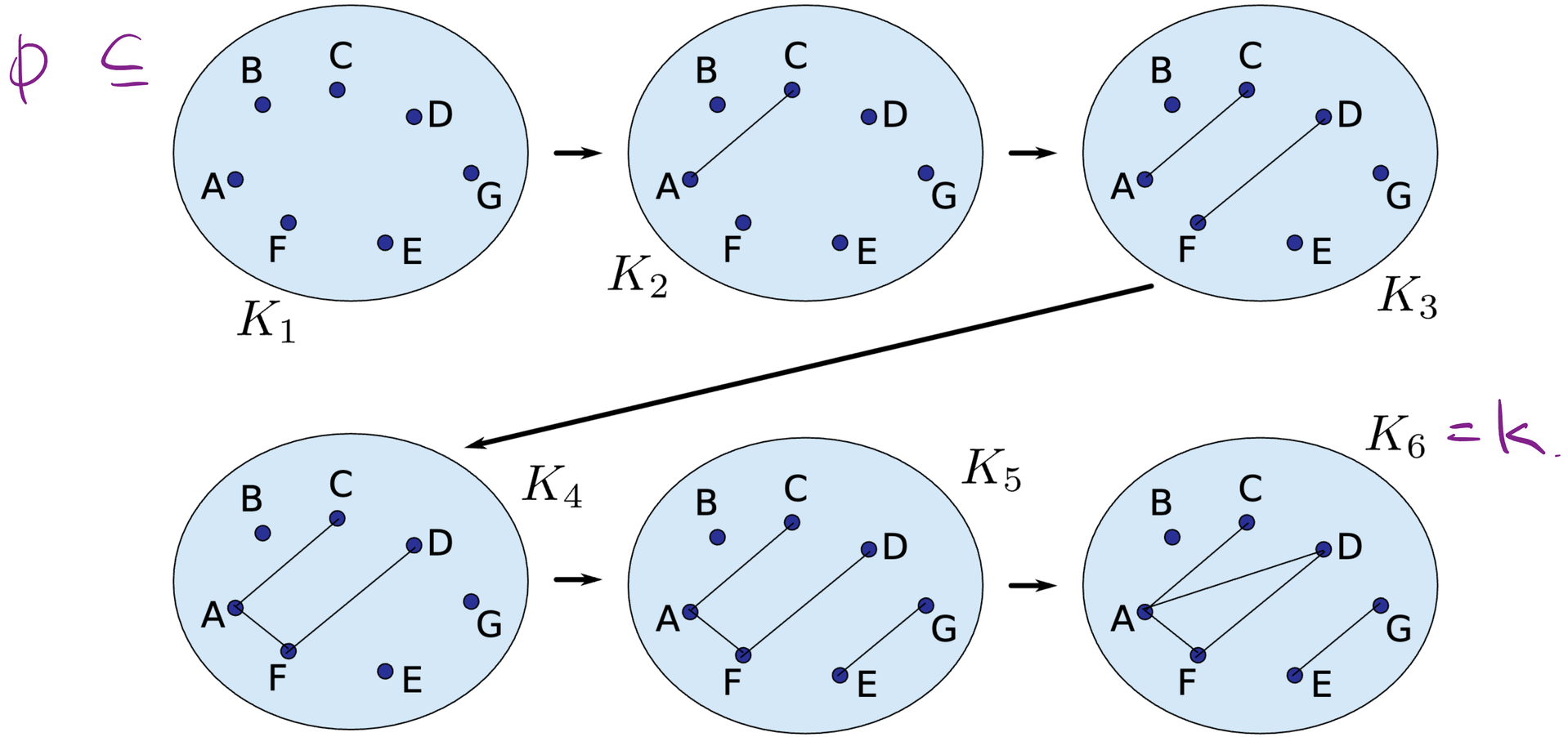
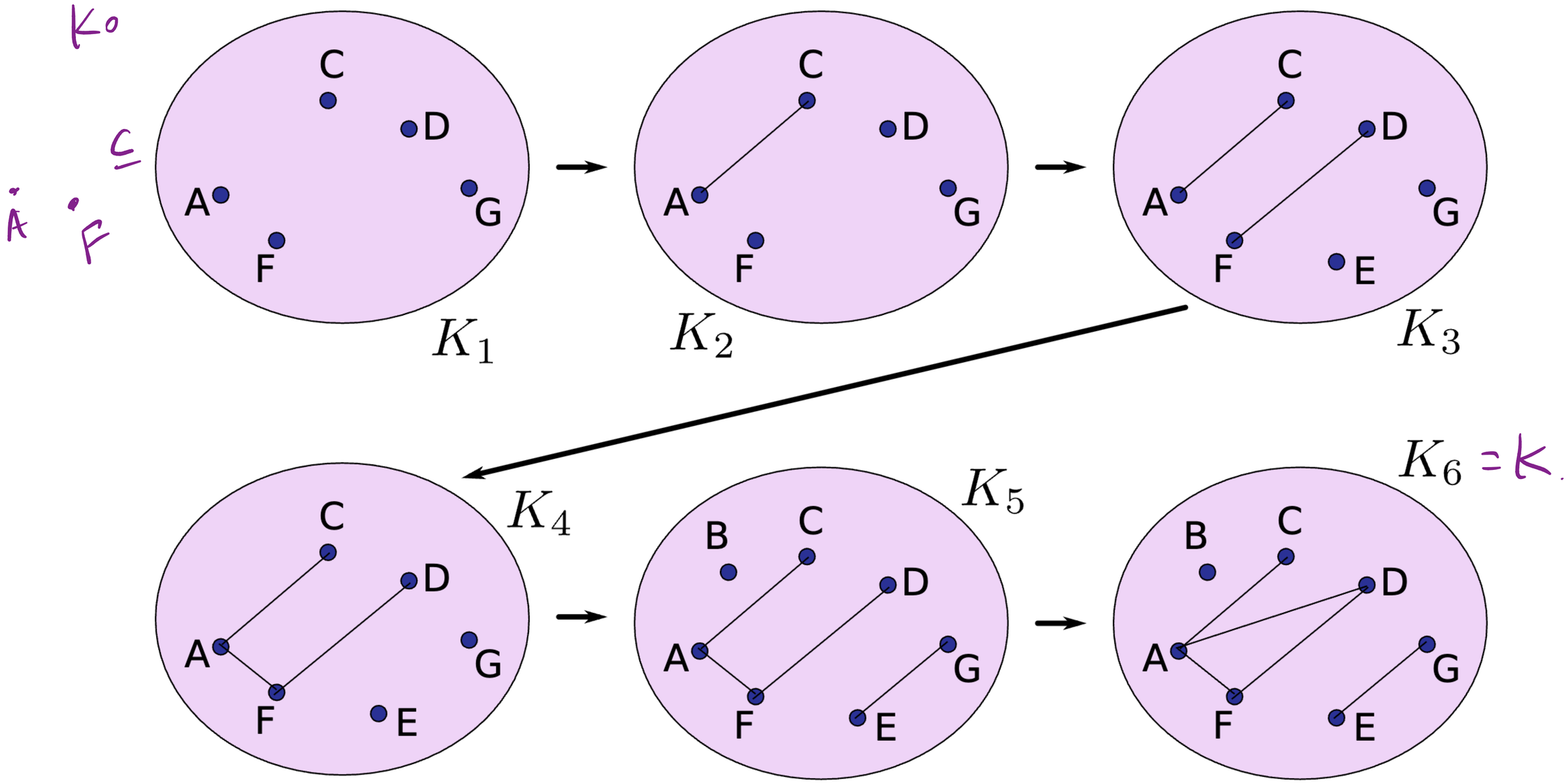


Example 2



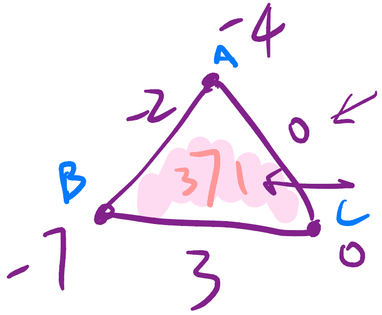
Monotone Function

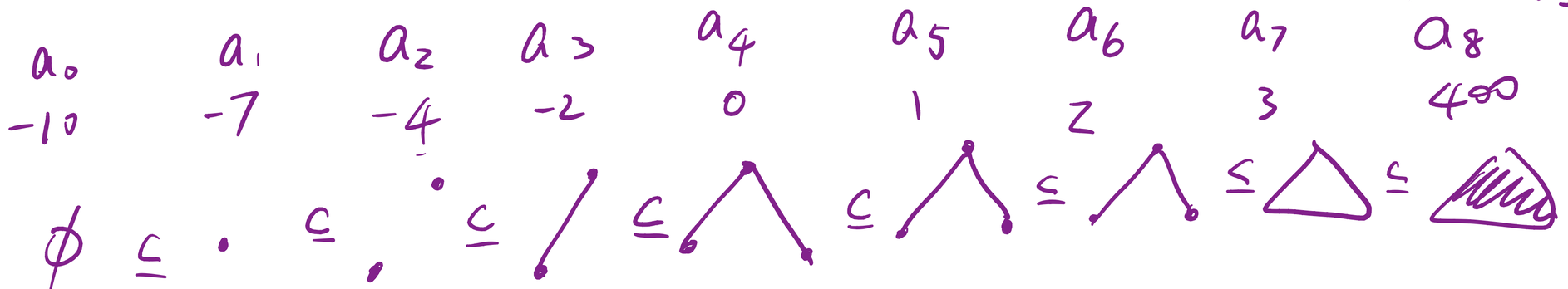
A function $f : K \to \mathbb{R}$ is called monotone if for every $\tau \leq \sigma, f(\tau) \leq f(\sigma)$.

A simplex-wise filtration means that exactly one simplex is added at each step.

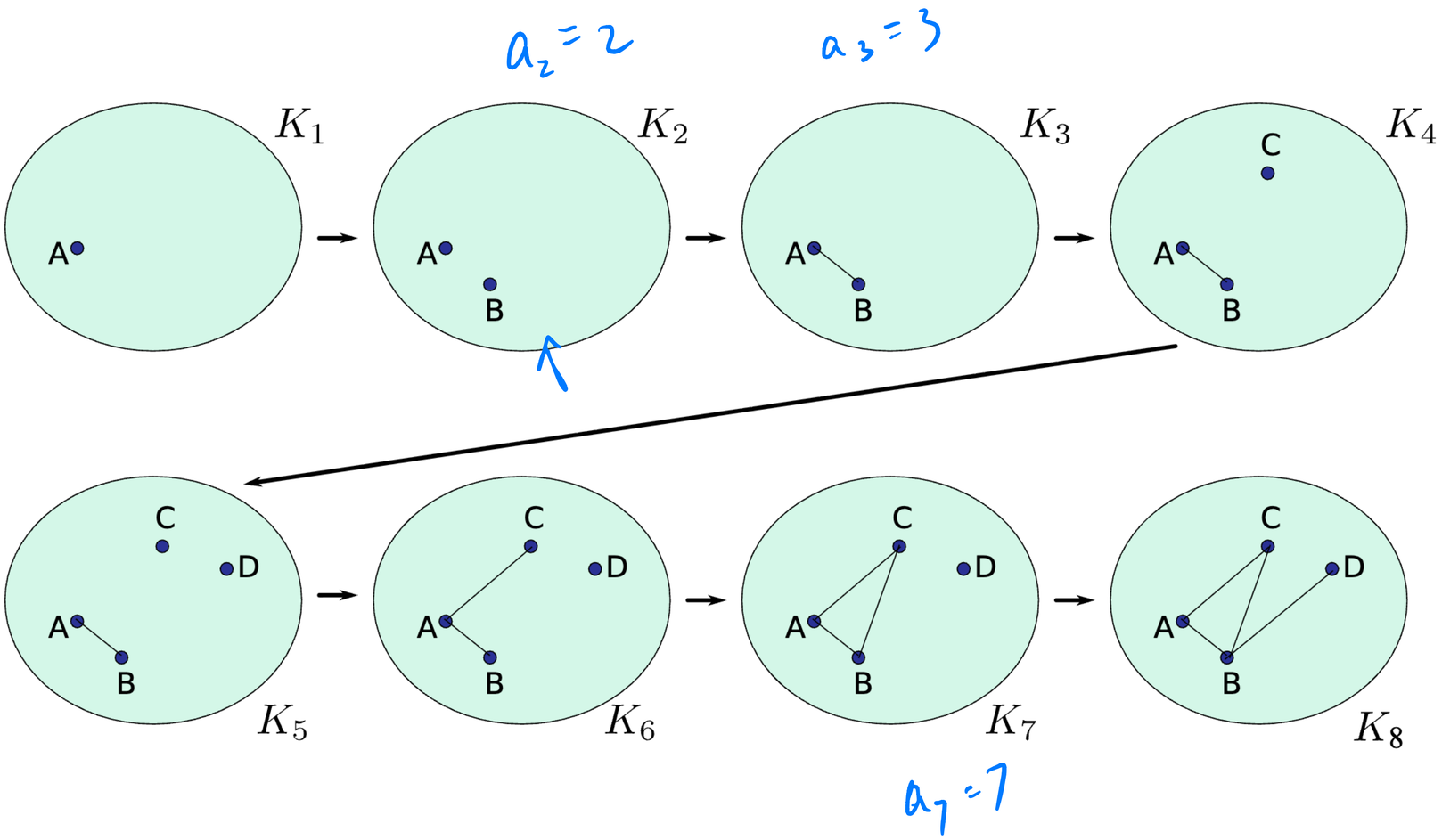
Fix values $a_0 \leq a_1 \leq \cdots \leq a_n$. Let $K_i = f^{-1}(-\infty, a_i]$. The sublevelset filtration induced by this function is defined to be

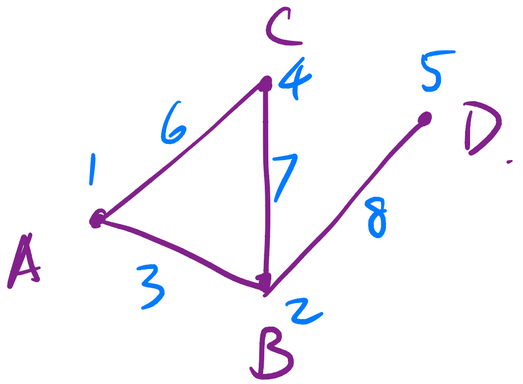
$$\emptyset = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_n = K.$$



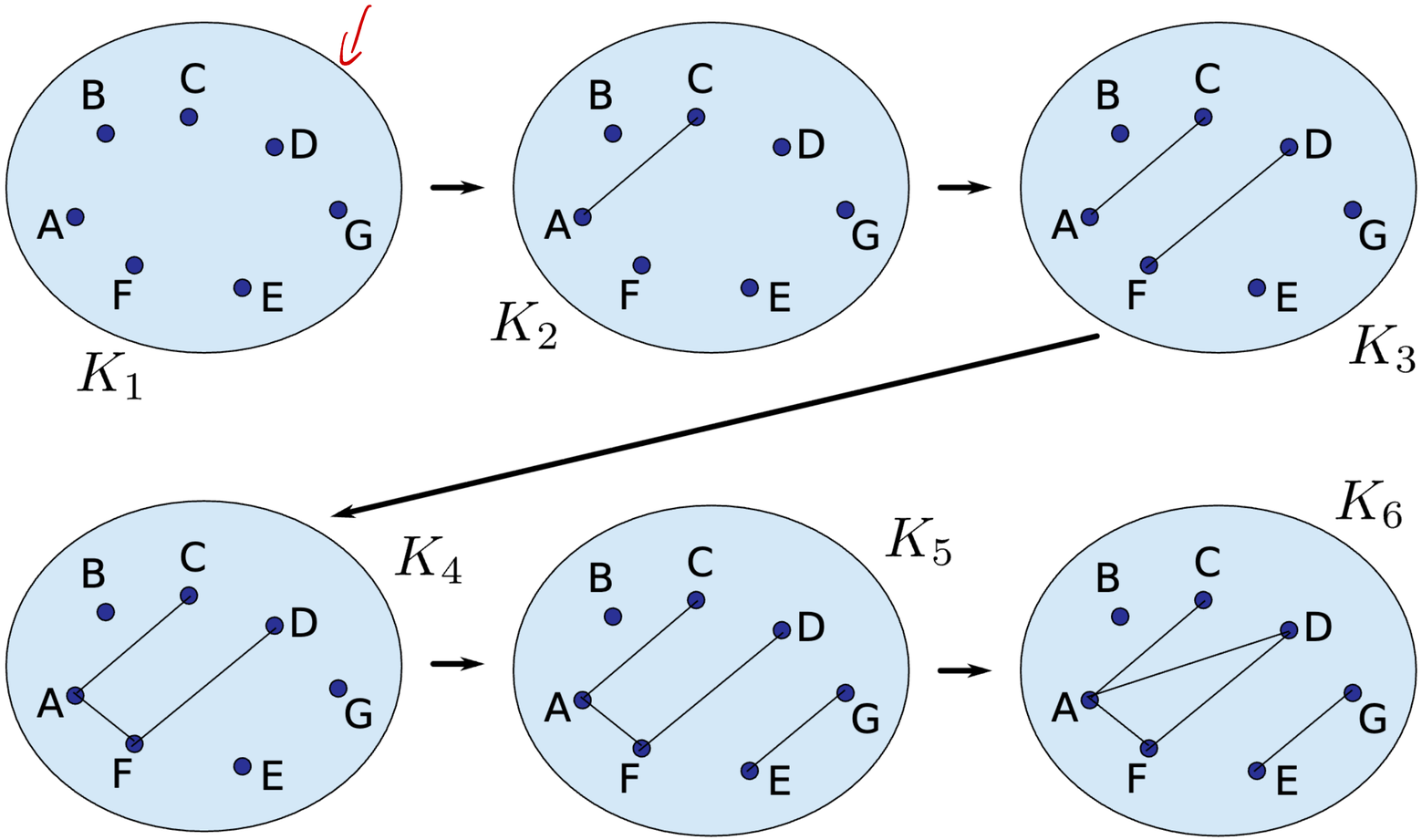


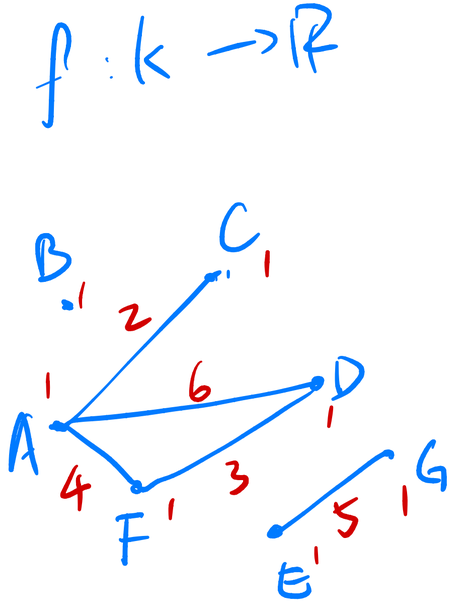
Example: Function inducing filtration





Try it: Function inducing filtration





Everything comes from a function

Lower Star Filtration

Given a height function $f : V(K) \to \mathbb{R}$ defined on vertices of K.

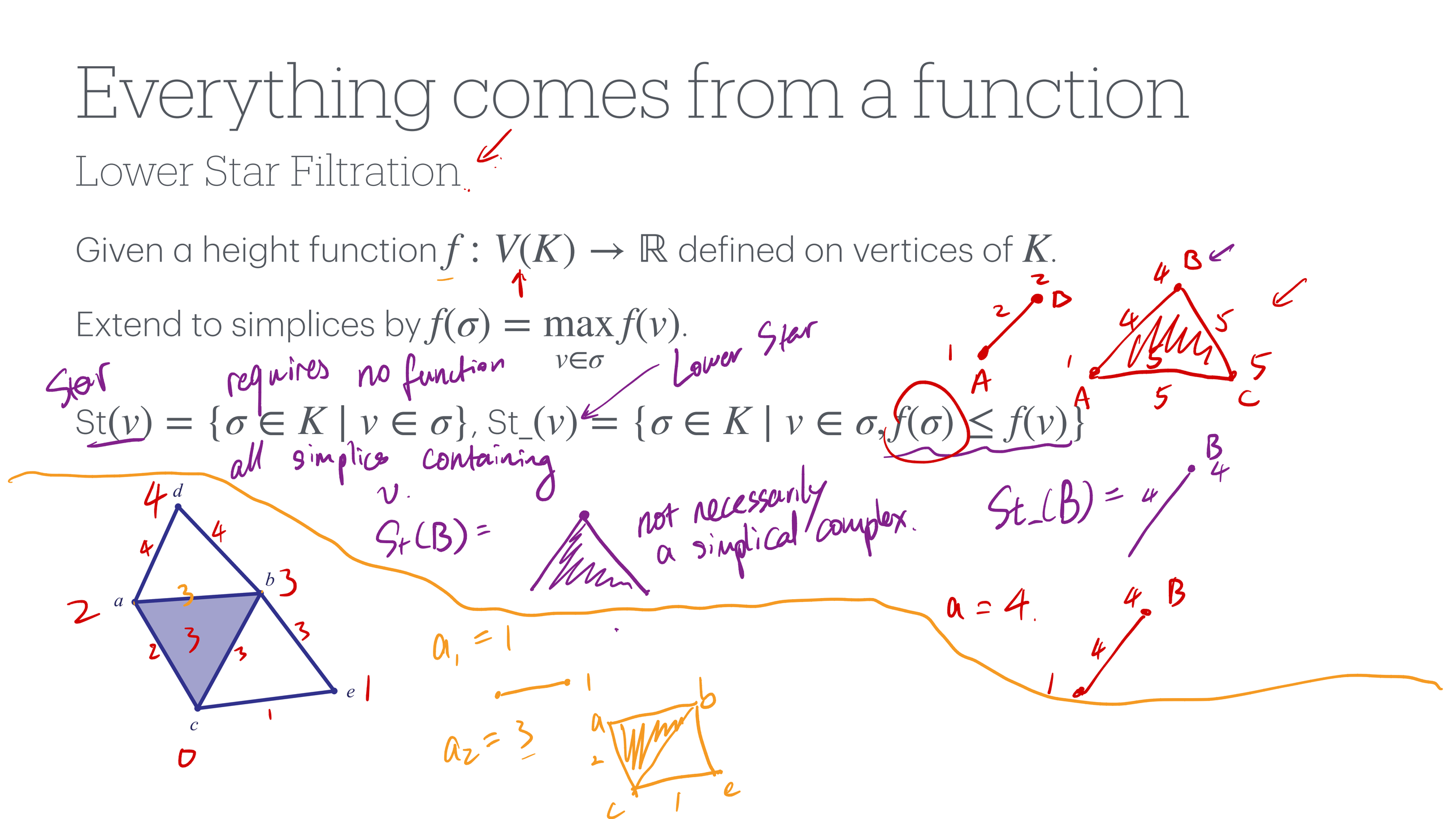
Extend to simplices by $f(\sigma) = \max_{v \in \sigma} f(v)$.

Star requires no function

Lower Star

$\mathrm{St}(v) = \{\sigma \in K \mid v \in \sigma\}$, $\mathrm{St}_{-}(v) = \{\sigma \in K \mid v \in \sigma, f(\sigma) \leq f(v)\}$

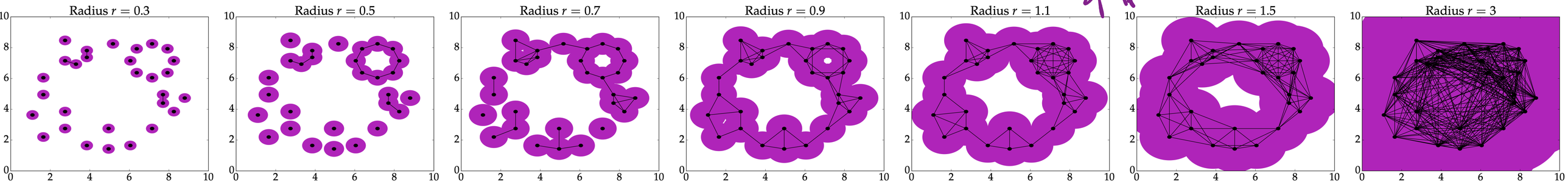
all simplices containing v.



Everything comes from a function

Rips complex filtration

$VR^r(P) = \{\sigma \in P \mid \|v-w\| \le 2r, \ \forall v,w \in \sigma\}$.



K : complete simplicial complex on N - vertice.

$r = 0$

$VP^0(P) = V$

$f : K \longrightarrow \mathbb{R}$

vertices $v \longmapsto 0 \quad \forall v \in V$.

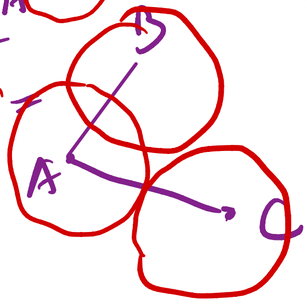
edge : $e \longmapsto \|v-w\|$.

higher dim: $\sigma \longmapsto \max_{v,w \in \sigma} \|v-w\|$.

$VR^0 = \{A, B, C\}$

$VR^{0.5} =$

$VR^{0.75} =$



$VR^{1.2} =$

What to do with a filtration?

Betti Curves

not continuous
not a curve

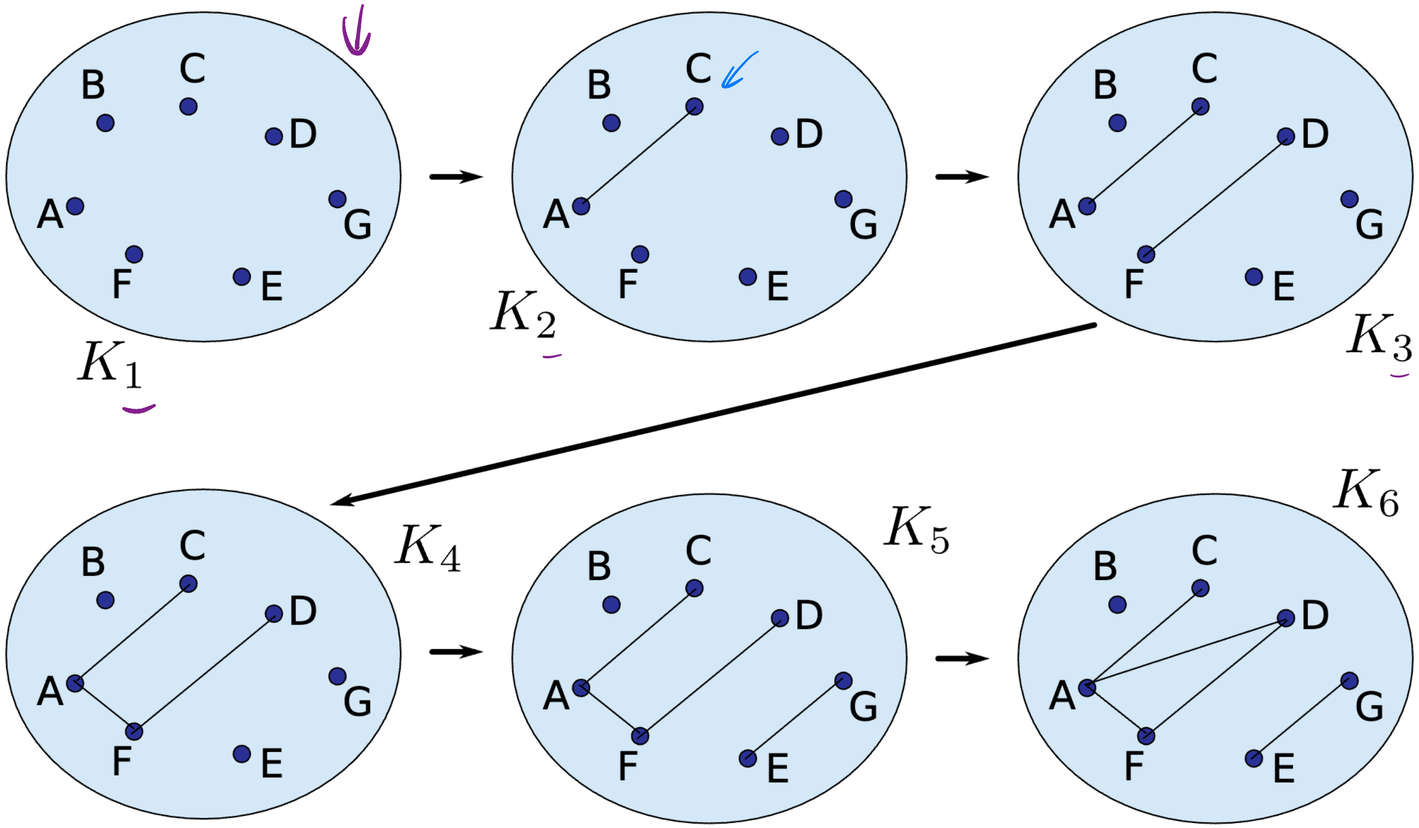
$\mathcal{F}: \emptyset \subseteq K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n$.
$-\infty < a_0 < a_1 < \cdots < a_n$. $a_i \in \mathbb{R}$

The p-th Betti curve is a function

$$\beta_p(\mathcal{F}) : \mathbb{R} \longrightarrow \mathbb{Z}$$
$$a_i \longmapsto \beta_p(K_i) = \dim(H_p(K_i)).$$

Example

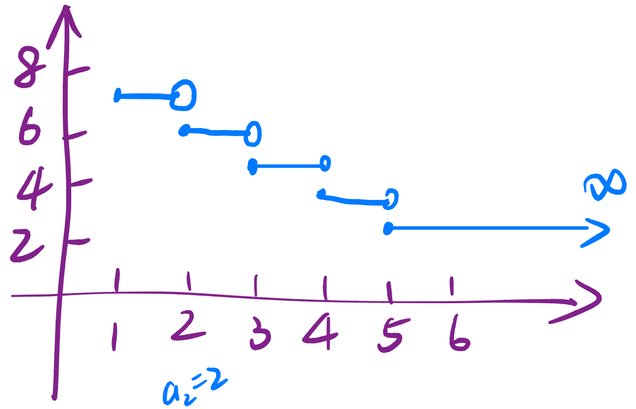
Write the **0**th and **1**st Betti curves for this filtration:



$\beta_0(F) = \beta_0(K_i)$

a_i

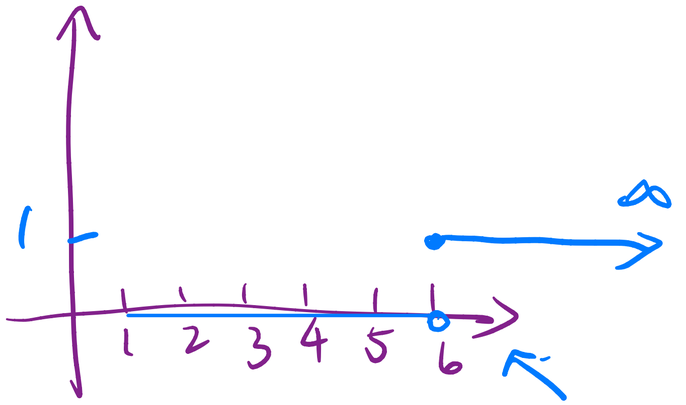
0-dim.



$a_i = 2$

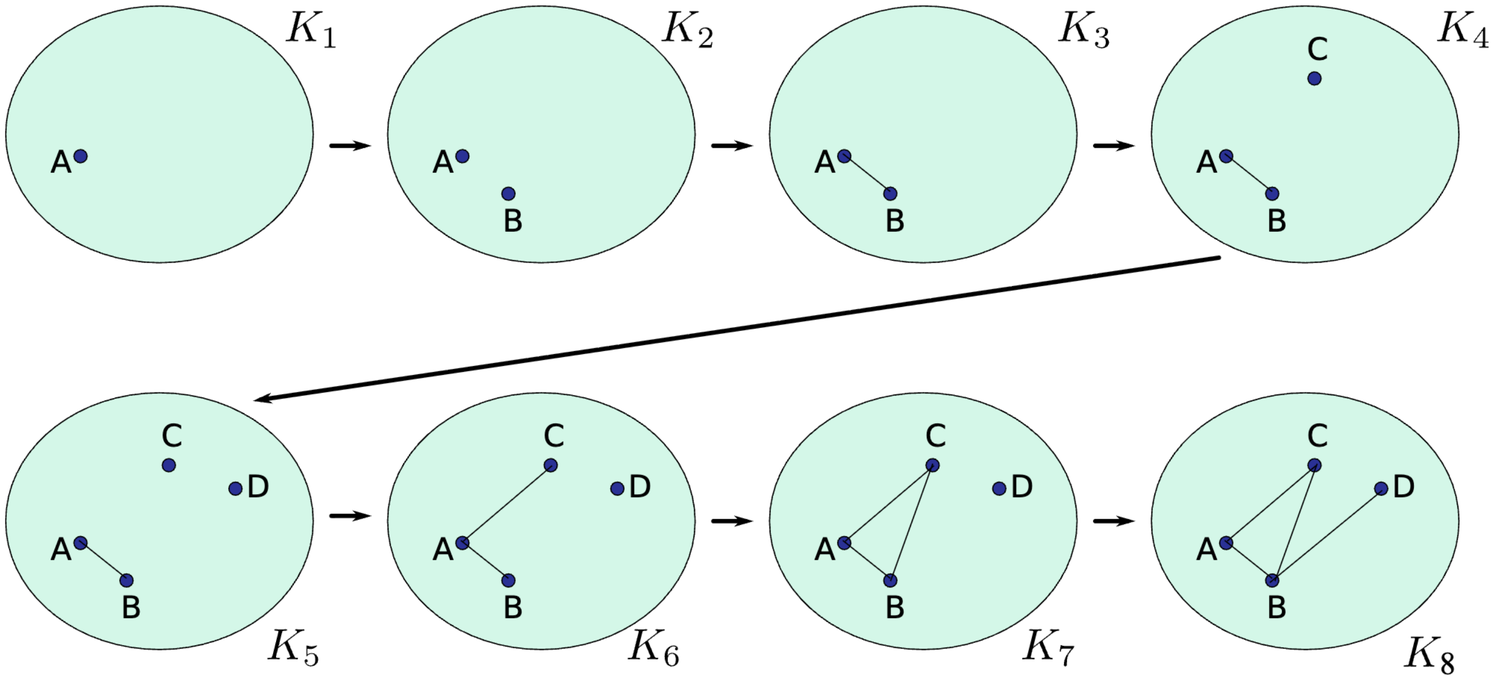
K_2.

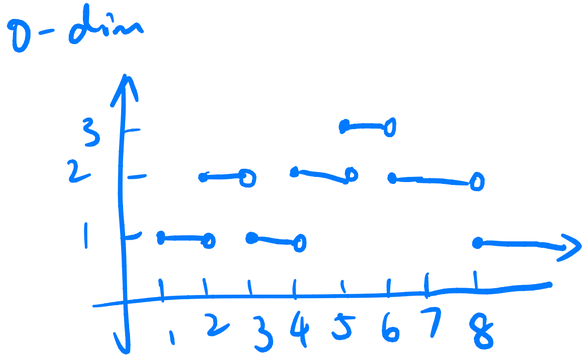
1-dim

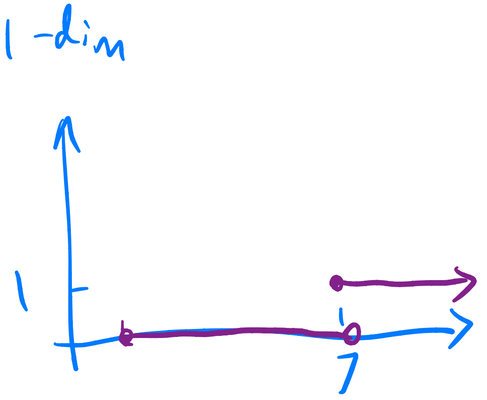


Try it

Write the **0**th and **1**st Betti curves for this filtration:





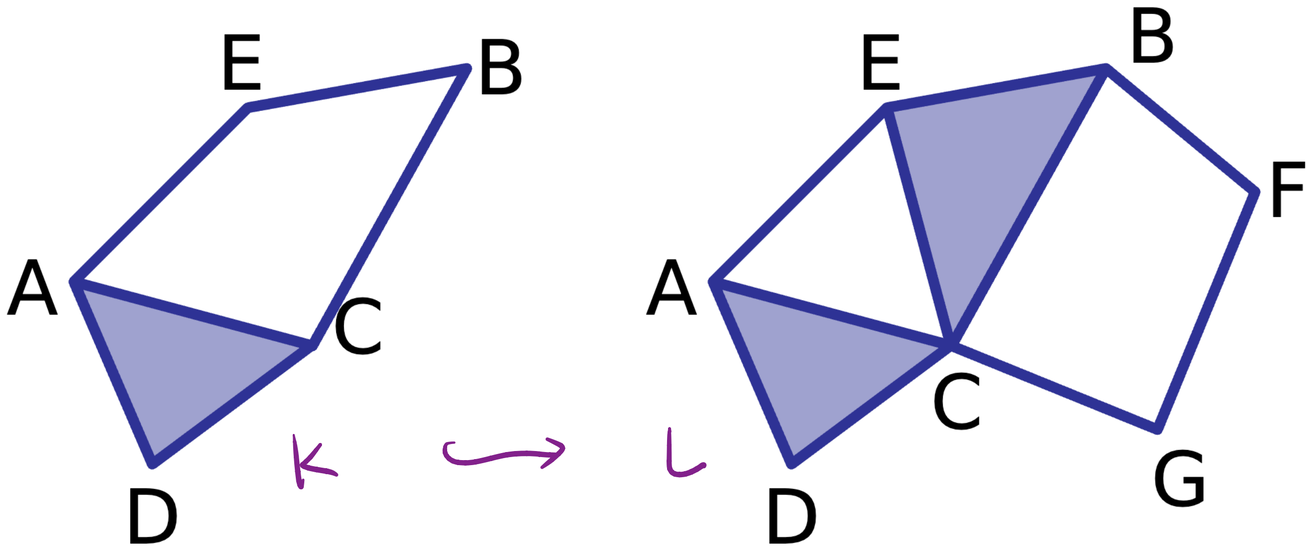


Induced Maps

Simplicial Maps

A *simplicial map* between abstract simplicial complexes $f : K \to L$ is induced by a map on the vertices $f : V(K) \to V(L)$. For every $\sigma \in K, f(\sigma) \in L$. ← No worries!

Cheap trick: we can ~~use~~ assume inclusion! ↩



$f(A) = A$

⋮

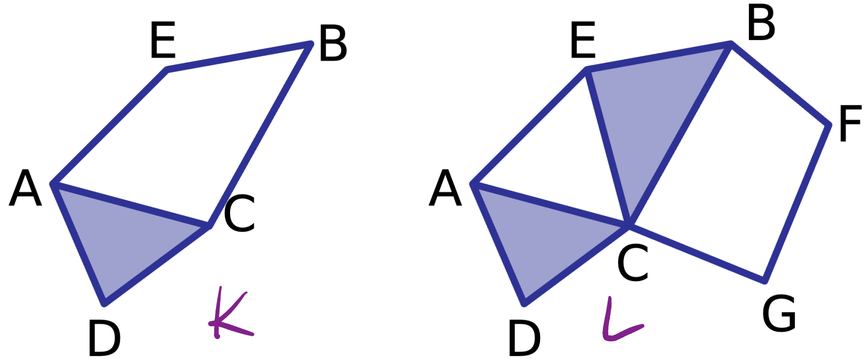
$f(E) = E$

$f(AE) = AE$

⋮

$f(ACD) = ACD$

Big Diagram of Vector Spaces



$$\cdots \longrightarrow C_2(K) \xrightarrow{\partial_2^K} C_1(K) \xrightarrow{\partial_1^K} C_0(K) \xrightarrow{\partial_0^K} 0$$

$$\downarrow f_\# \qquad \downarrow f_\# \qquad \downarrow f_\#$$

$$\cdots \longrightarrow C_2(L) \xrightarrow{\partial_2^L} C_1(L) \xrightarrow{\partial_1^L} C_0(L) \xrightarrow{\partial_0^L} 0$$

AC+CD+AD

AC+CD+AD.

Simplicial map. $f : K \hookrightarrow L$.

$f(A) = A$

$\vdots$

$\Downarrow$

linear maps between chain groups

$f_\# : C_p(K) \to C_p(L)$.

f: acts on simplices.

$f_\#$: acts on chains.

f_*: acts on homology classes.

Induced Map

[boundary] = [0]

$H_1 = Z_1 / B_1$

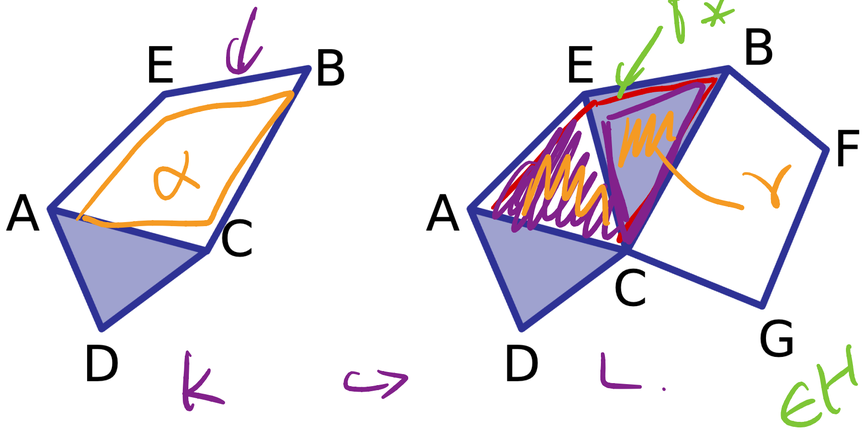
Given a simplicial map $f : K \to L$, the induced map on homology is defined by

$$f_* : H_p(K) \longrightarrow H_p(L)$$
$$[\alpha] \longmapsto [f_\#(\alpha)]$$

a hole?

hole!

$H_1(K)$



$f_*([\alpha])$

α: a cycle.

$[\alpha]$: homology class represented by α.

$f_\#(\alpha)$: image of α.

$[f_\#(\alpha)]$: homology class of that image.

$\in H_1(L)$?

$f_*([\alpha]) = [AE + EB + BC + CA]$

$\partial_2(BCE) = BC + CE + BE$

$AE + AC + EC = f_*([\alpha]) + \partial_2(BCE) + \partial_2(AEC)$

in $H_1(L)$.

$[f_*(\alpha)] = [\beta] \Rightarrow -f_*(\alpha) \neq 0$

$f_*(\alpha) = AE + EB + AC + CB$

$= \partial_2(AEC) + \partial_2(EBC)$

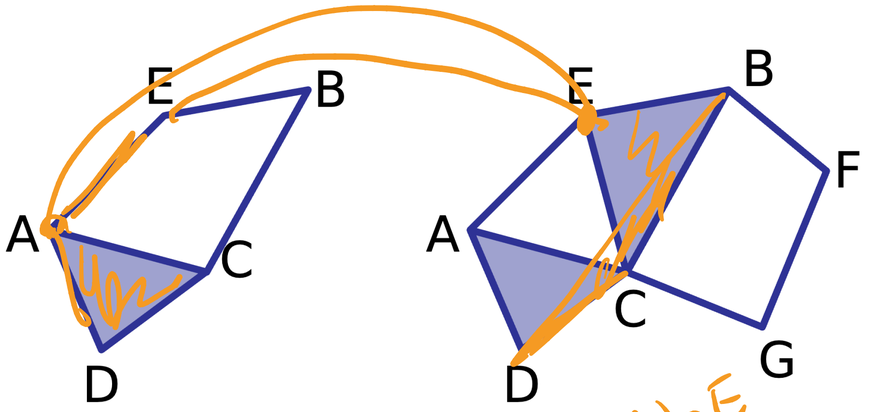
$= [0] + [0]$

$= [0]$

$f_*([\alpha]) = 0$

Commutative Diagram

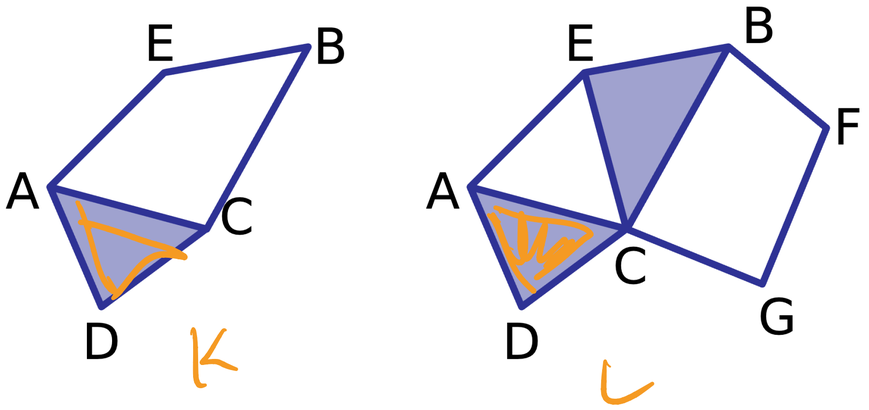
A diagram commutes if any path of functions is equal.



$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C_2(K) & \xrightarrow{\partial_2^K} & C_1(K) & \xrightarrow{\partial_1^K} & C_0(K) & \xrightarrow{\partial_0^K} & 0 \\
 & & \downarrow f_\# & & \downarrow f_\# & & \downarrow f_\# & & \\
\cdots & \longrightarrow & C_2(L) & \xrightarrow{\partial_2^L} & C_1(L) & \xrightarrow{\partial_1^L} & C_0(L) & \xrightarrow{\partial_0^L} & 0
\end{array}$$

Claim: $f_\# \circ \partial^K = \partial^L \circ f_\#$.

Effect on Cycles

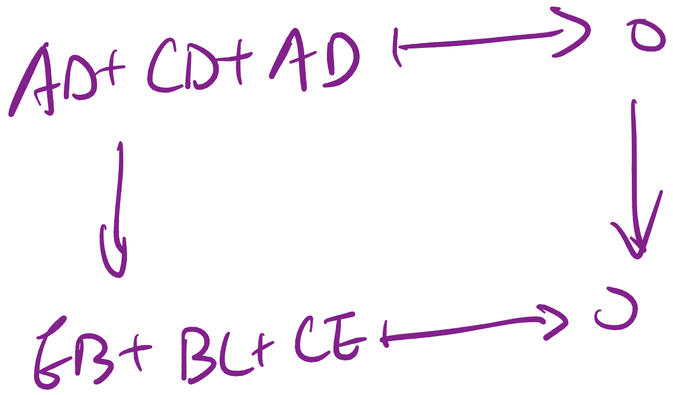


Cycles go to cycles!

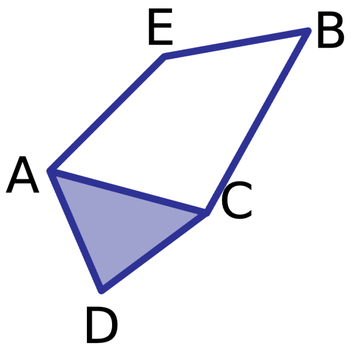
$$\begin{array}{ccccccccc} \cdots & \longrightarrow & C_2(K) & \xrightarrow{\partial_2^K} & C_1(K) & \xrightarrow{\partial_1^K} & C_0(K) & \xrightarrow{\partial_0^K} & 0 \\ & & \downarrow f_\# & & \downarrow f_\# & & \downarrow f_\# & & \\ \cdots & \longrightarrow & C_2(L) & \xrightarrow{\partial_2^L} & C_1(L) & \xrightarrow{\partial_1^L} & C_0(L) & \xrightarrow{\partial_0^L} & 0 \end{array}$$

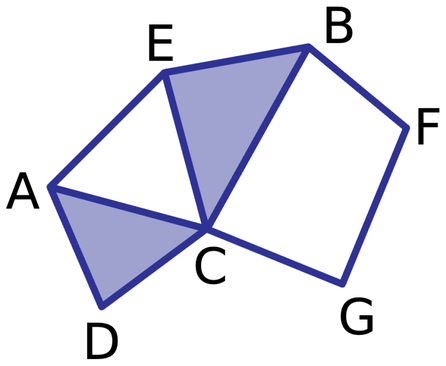
AD + CD + AD $\longmapsto$ 0

AD + CD + AD $\longmapsto$ 0



Effect on Boundaries





Boundaries go to boundaries!

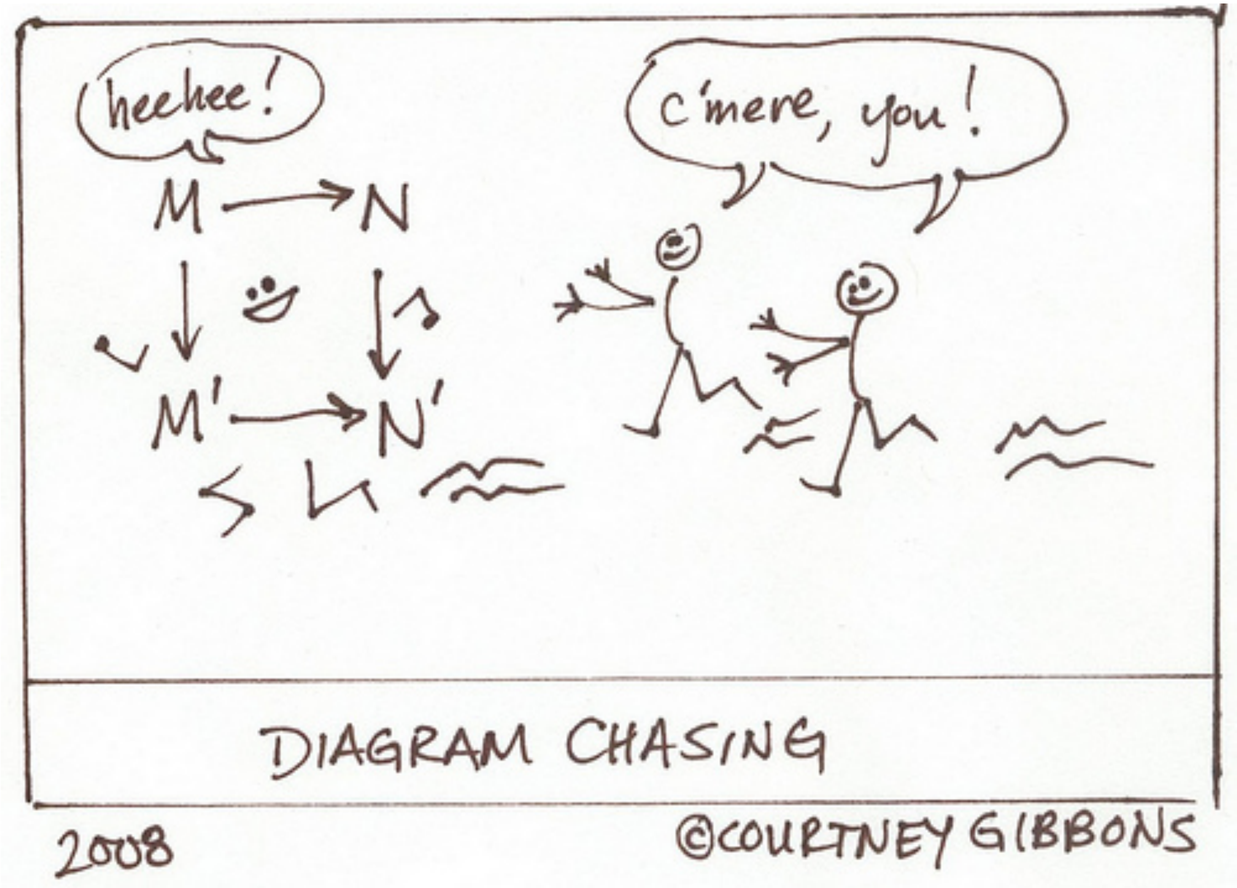
$$\begin{array}{ccccccccc} \cdots & \longrightarrow & C_2(K) & \xrightarrow{\partial_2^K} & C_1(K) & \xrightarrow{\partial_1^K} & C_0(K) & \xrightarrow{\partial_0^K} & 0 \\ & & \downarrow f_\# & & \downarrow f_\# & & \downarrow f_\# & & \\ \cdots & \longrightarrow & C_2(L) & \xrightarrow{\partial_2^L} & C_1(L) & \xrightarrow{\partial_1^L} & C_0(L) & \xrightarrow{\partial_0^L} & 0 \end{array}$$

$ACD \longrightarrow AD+CD+AC$

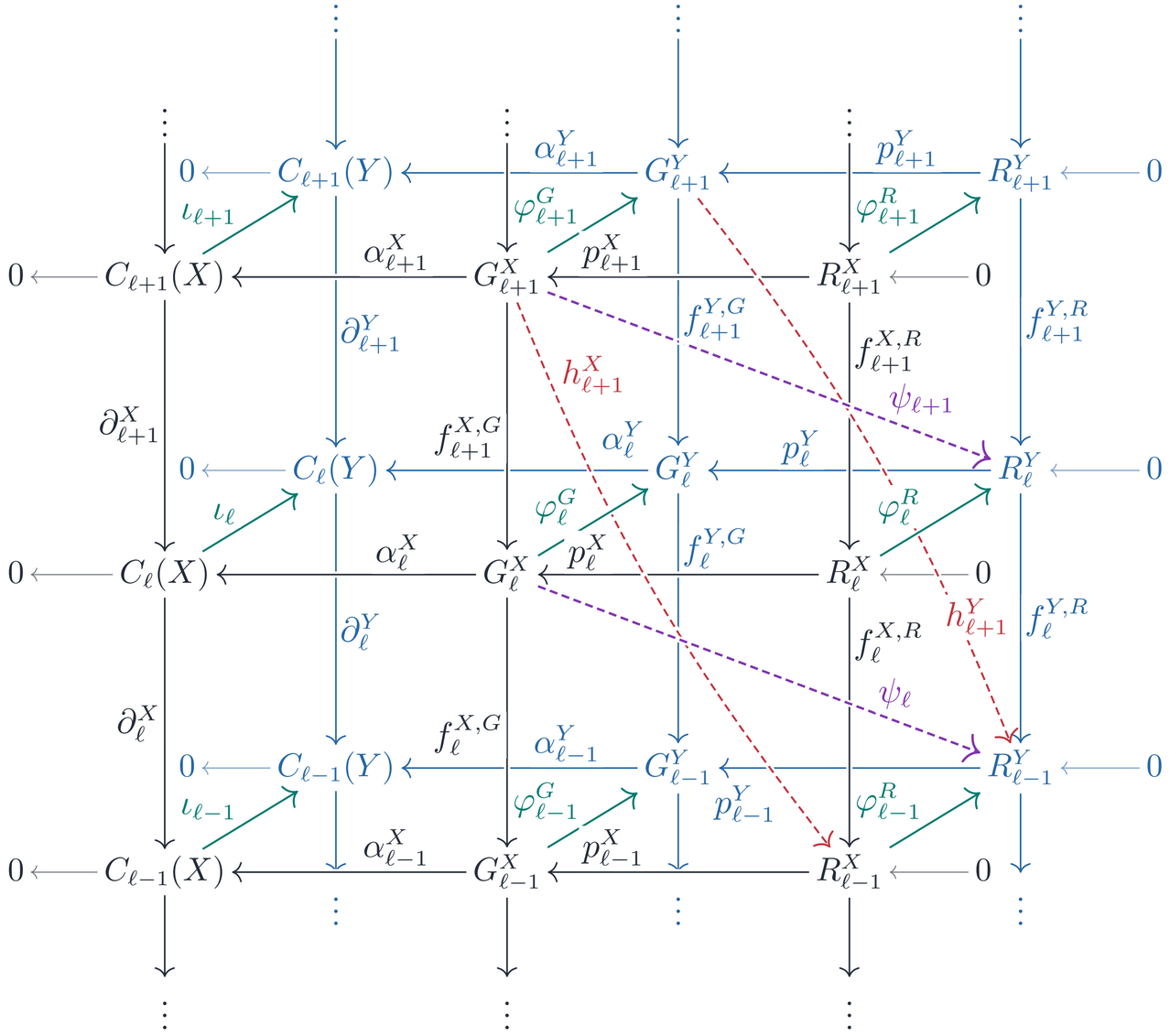
$ACD \longrightarrow AD+CD+AC$

$ECB \longrightarrow EB+BL+CE.$

Diagram Chasing



Sometimes diagrams are scary...



Credit: Florian Russold

Example

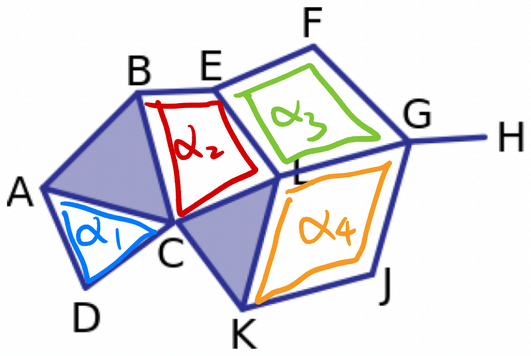
If $H_1(K)$ generated by

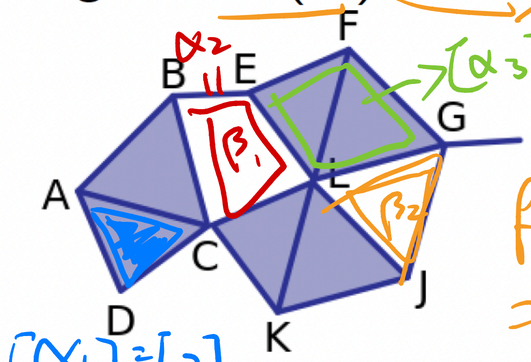
$$[\alpha_1] = [AC + CD + AD]$$
$$[\alpha_2] = [BE + EL + CL + CB]$$
$$[\alpha_3] = [EF + FG + GL + EL]$$
$$[\alpha_4] = [GL + GJ + JK + LK]$$

and $H_1(L)$ generated by

$$[\beta_1] = [BE + EL + CL + CB]$$
$$[\beta_2] = [GL + GJ + LJ]$$

write the matrix representing $f_* : H_1(K) \to H_1(L)$ induced by inclusion.





hole — a hole ?

$\alpha_2 = \beta_1$

$[\alpha_3] = [0]$

$f_*([\alpha_4]) = [\beta_2]$

$[\alpha_1] = [0]$

	α_1	α_2	α_3	α_4
β_1	0	1	0	0
β_2	0	0	0	1

Homework Presentation 1

Loris.

- Let $K = \{A, B, C, D, AB, BC, AC, CD, AD, DE, AE\}$, and L be obtained from K by adding ABC. Suppose $H_1(K)$ has basis
  $\alpha_1 = AB + BC + AC$, $\alpha_2 = AC + CD + AD$, $\alpha_3 = AD + DE + AE$ and $H_1(L)$ has basis
  $\beta_1 = AC + CD + DE + AE$, $\beta_2 = AD + DE + AE$.
  - Draw K and L.
  - Which of the three homology classes in $H_1(K)$ becomes trivial in $H_1(L)$? Why?
  - Write the matrix of the induced map $f_* : H_1(K) \longrightarrow H_1(L)$.
  - Is f_* injective? Is it surjective? Explain geometrically

Homework Presentation 2 pengdeng

- Let $K = \{A, B, C, D, AB, BC, AC, CD, AD\}$, and L be obtained from K by adding ABC, two vertices E and F, and the edges AE, EF, FA . Suppose $H_1(K)$ has basis $\alpha_1 = AB + BC + AC$, $\alpha_2 = AC + CD + AD$ and $H_1(L)$ has basis $\beta_1 = AC + CD + AD$, $\beta_2 = AF + FE + AE$.
  - Draw K and L.
  - Compute $f_*([\alpha_1])$ and $f_*([\alpha_2])$, and write the matrix representing $f_* : H_1(K) \longrightarrow H_1(L)$ with respect to the given bases.
  - Notice that $\beta_1(K) = \beta_1(L) = 2$. Does this mean both holes in K survive in L?
  - Identify a nonzero homology class in $H_1(K)$ that maps to zero in $H_1(L)$. Explain geometrically why this happens.

Midsemester Evaluation: